

Risk-Informed Adaptive Construction of Enhanced Bayesian Networks: Sampling, Variable Discretization, and CPT Learning for System Reliability

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1. Introduction

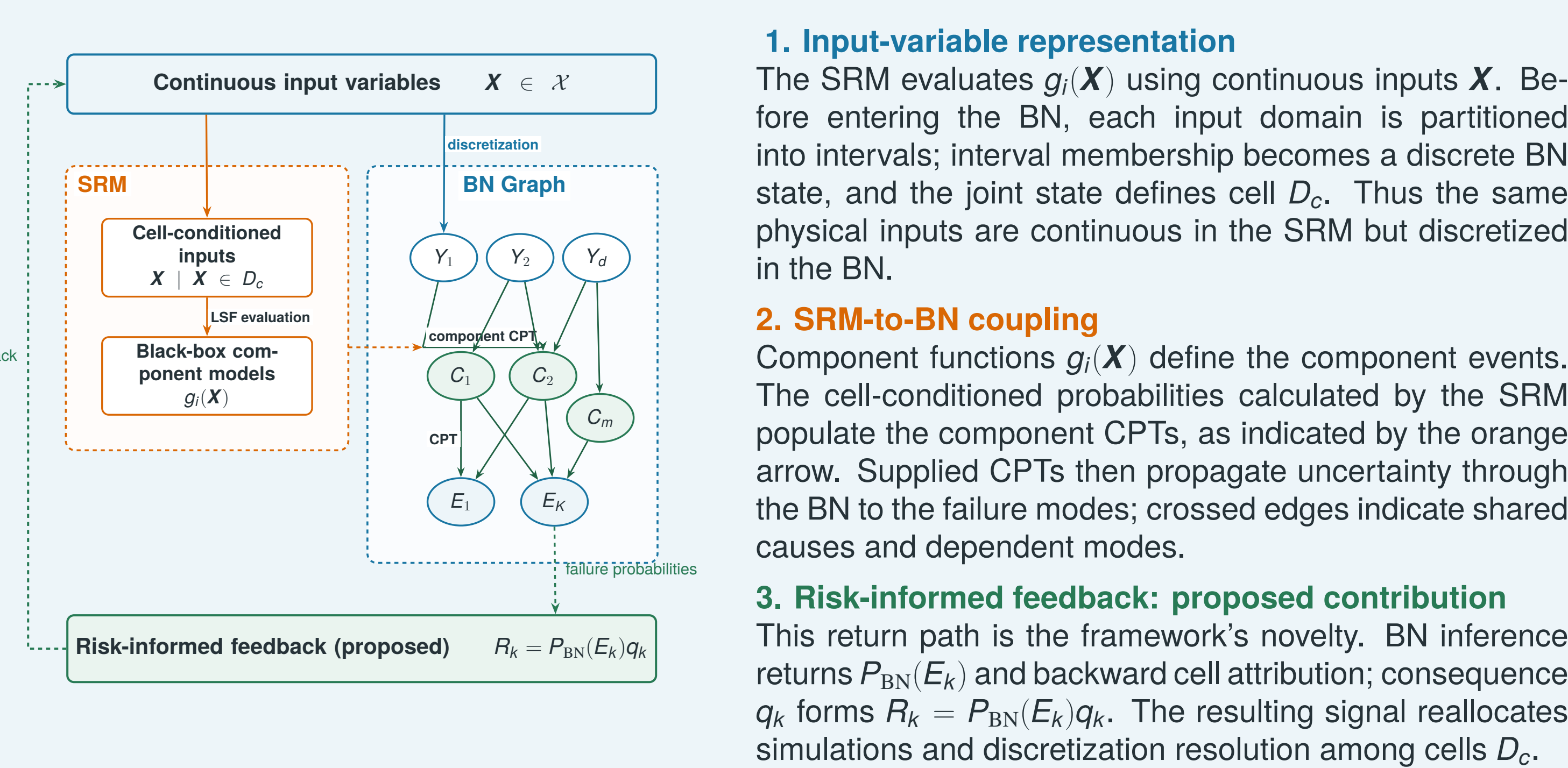
A Bayesian network (BN) represents random variables as nodes and their conditional dependence as directed edges. For a discrete BN, each child node is quantified by a **conditional probability table (CPT)**, which gives the probability of every child state for each combination of its parent states. The discrete BN used here therefore requires continuous engineering variables to be replaced by a finite set of states before standard BN inference can be applied.

An **enhanced Bayesian network (eBN)** combines a continuous **structural reliability method (SRM)** with this discrete probabilistic representation (Straub and Der Kiureghian, 2010a). Limit-state functions $g_i(\mathbf{X})$ define component events $C_i = \{g_i(\mathbf{X}) \leq 0\}$ under shared uncertain inputs $\mathbf{X}$. The SRM supplies the conditional event probabilities needed for the component CPTs, after which the model is reduced to a discrete BN for forward and backward inference. In nonlinear or black-box models, these probabilities must often be estimated through costly simulations. Accurate reduction then requires both suitable state boundaries and enough evaluations in each state: coarse discretization loses information, whereas fine discretization spreads a limited simulation budget over many CPT entries. We therefore seek the coarsest discretization that preserves the accuracy needed for downstream BN inference.

Constructing such an eBN therefore couples two expensive decisions: where to evaluate the black-box component models and where to place the discrete BN state boundaries. Existing rare-event discretization and surrogate-assisted CPT-learning approaches do not generally co-adapt both decisions using downstream consequences across multiple target events.

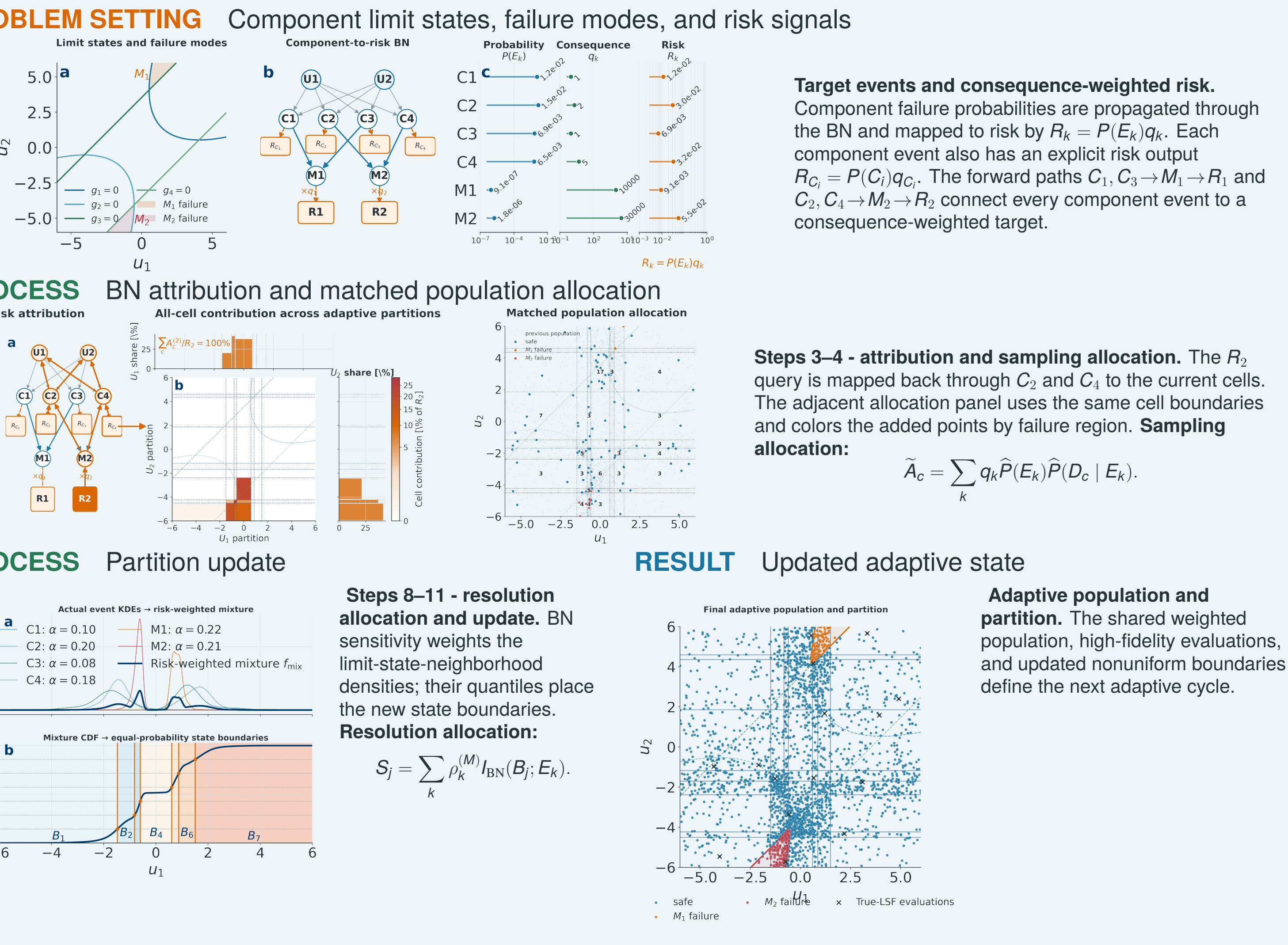
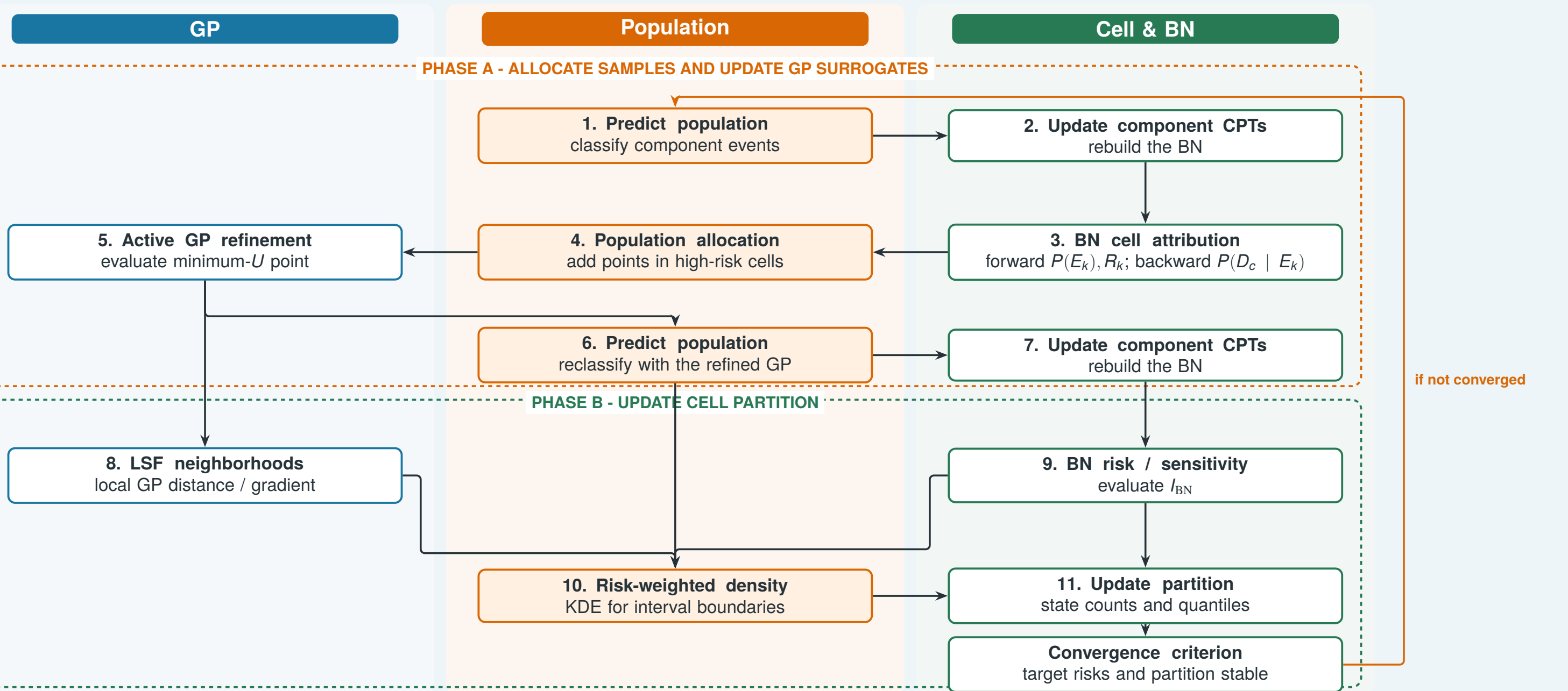
We therefore propose a **risk-informed computational interface** in which a shared weighted population and Gaussian-process (GP) surrogates estimate component CPTs, while $R_k = P_{\text{BN}}(E_k)q_k$ guides both high-fidelity sampling and input-state resolution across rare, unequal-consequence events.

2. Framework



3. Risk-Informed Sampling and Discretization

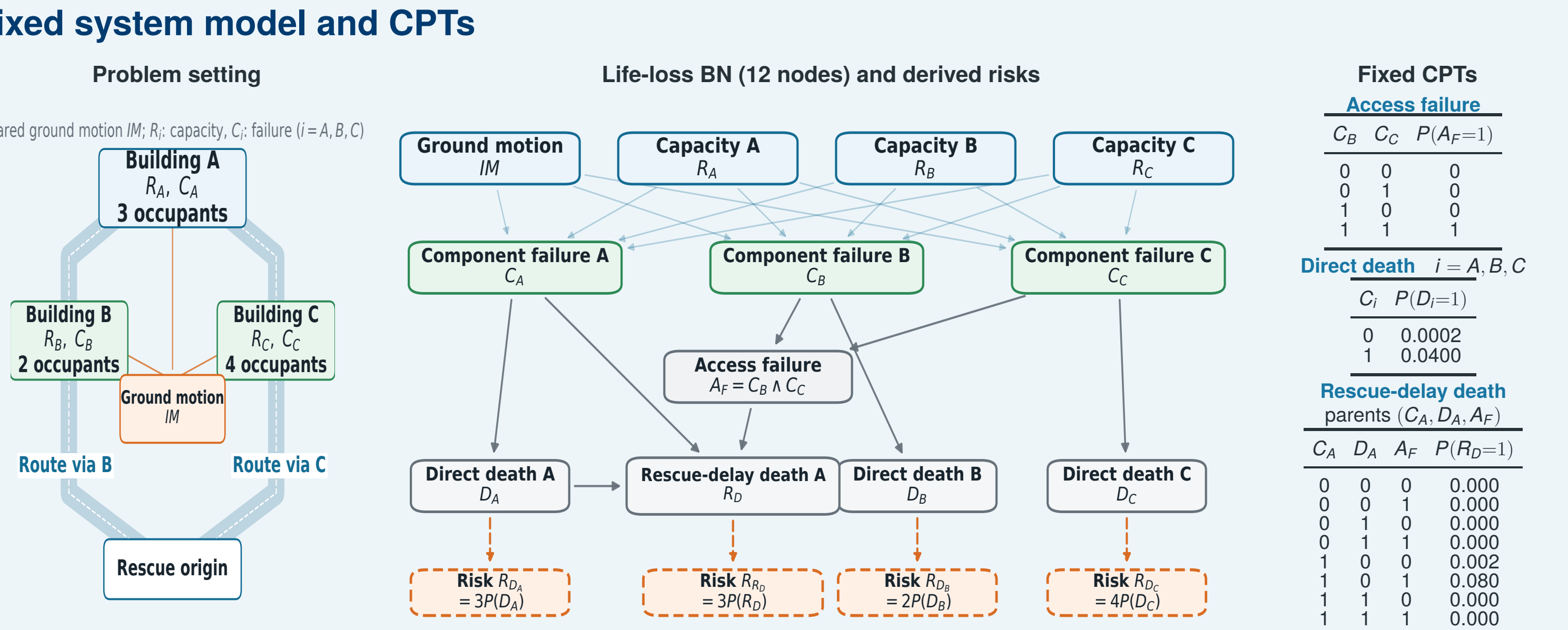
The workflow synchronizes the **GP**, **Population**, and **Cell & BN** lanes. Steps 1–5 predict component events, rebuild CPTs, attribute risk to cells, allocate samples, and refine the surrogates; Steps 6–7 refresh the predictions and CPTs. Steps 8–11 combine limit-state neighborhoods, BN sensitivity, and the risk-weighted population to update the state boundaries. Unlike earlier limit-state-surface-based discretization (Fukuda et al., 2026), this loop co-adapts sampling and resolution using consequence-weighted risk across multiple events.



4. Application

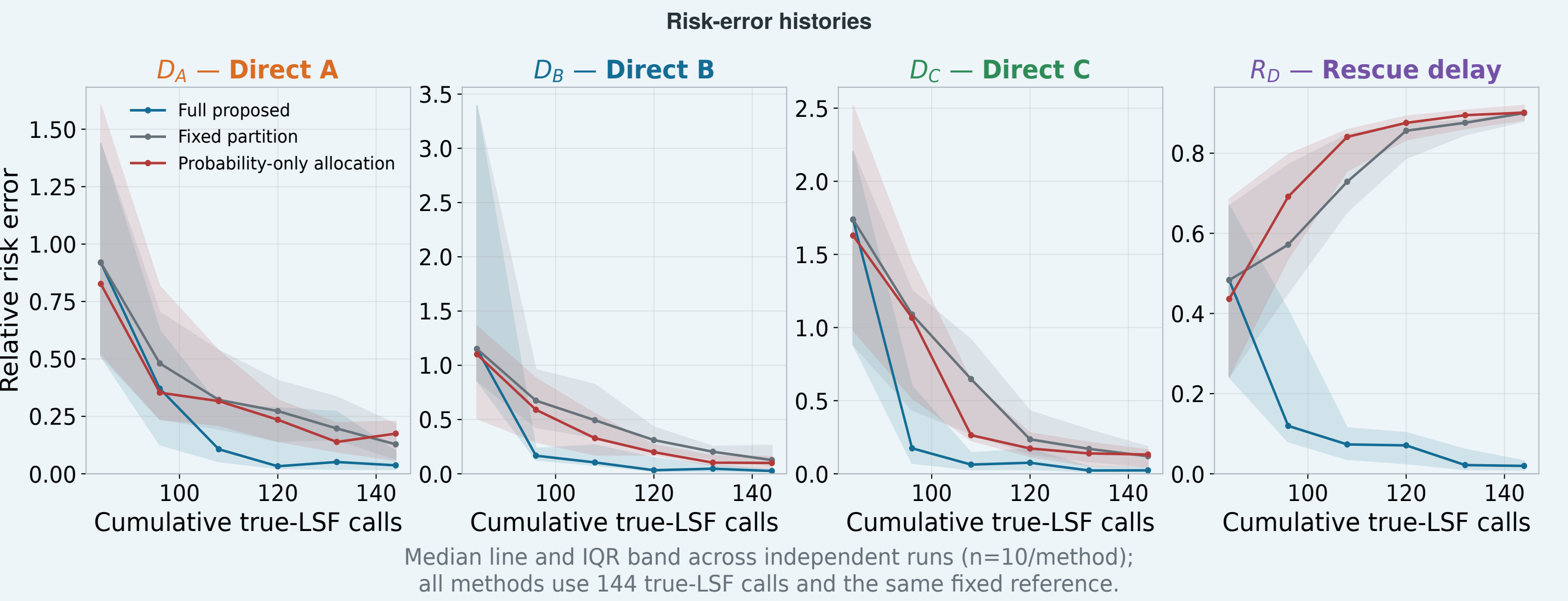
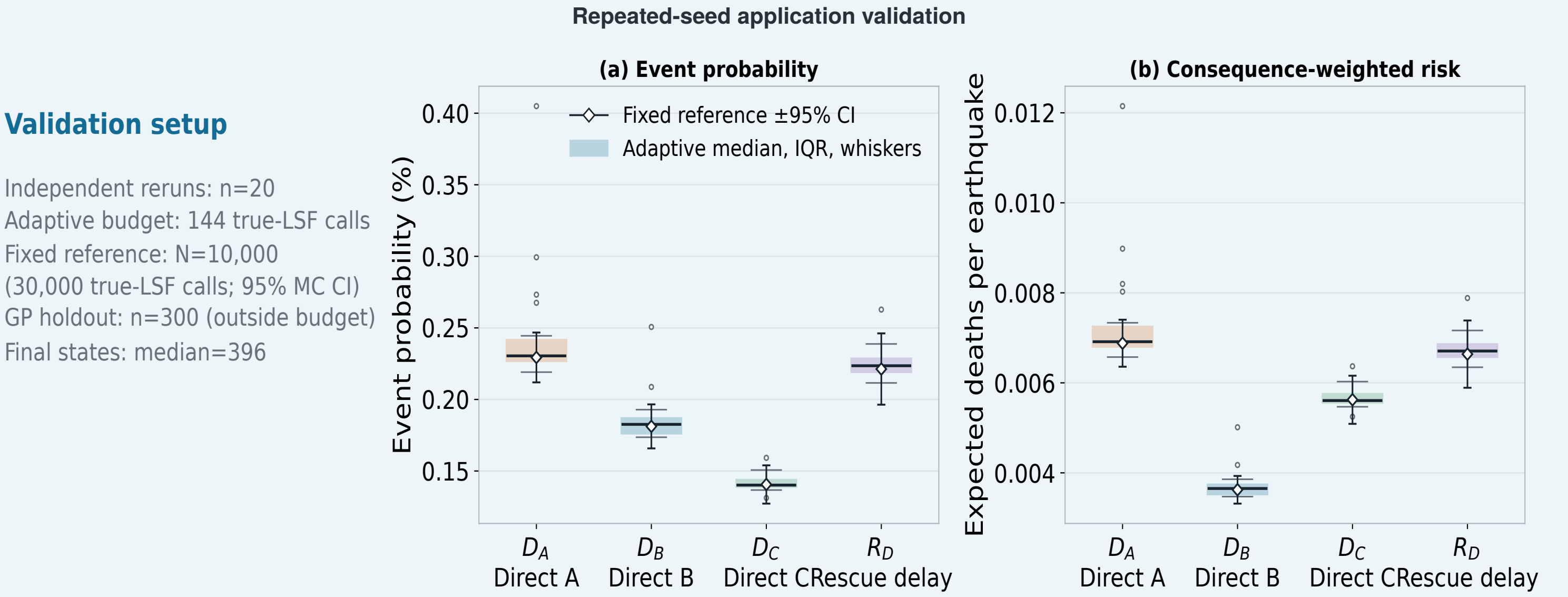
Why road blockage matters. Some Japanese cities contain densely built residential districts threaded by narrow streets. During strong shaking, collapsed buildings and debris can obstruct evacuation and emergency access; road blockage is therefore a system-level life-safety pathway, not only a transport detail (MLIT, 2006).

Minimum implementation. This deliberately small example communicates the framework rather than a calibrated urban model. Buildings A–C share ground-motion intensity IM , while R_A, R_B, R_C describe uncertain capacities. Rescue teams can reach A through a route past B or a route past C; access fails only when both $C_B = 1$ and $C_C = 1$. The 12-node BN estimates direct-collapse death at A/B/C and access-dependent rescue-delay death at A. Dashed orange nodes then convert these BN event probabilities into consequence-weighted risks; they are derived outputs rather than additional BN variables.



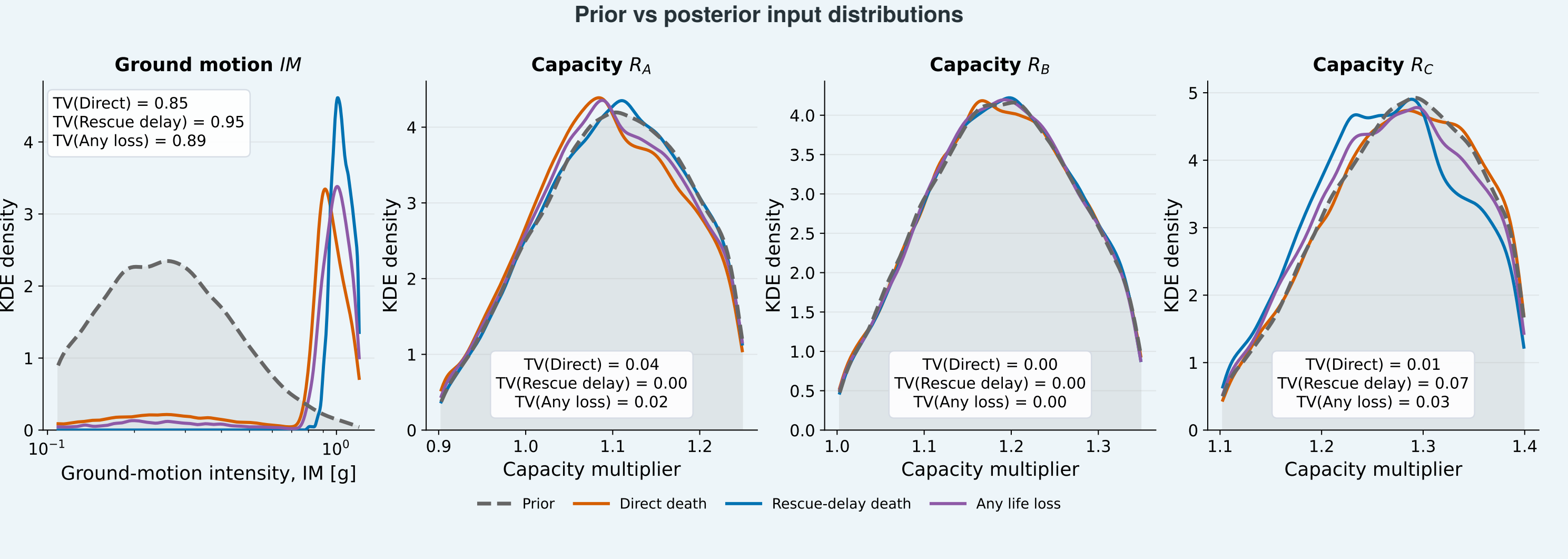
Validation

Repeated-seed application validation. The complete adaptive pipeline was rerun independently ($n = 20$) under a common 144-call true-LSF budget and compared with a fixed 10,000-sample reference (30,000 true-LSF calls, 95% interval). The common 300-sample GP holdout is excluded from the adaptive budget.



Sensitivity analysis

Each continuous curve is reconstructed by sampling from the prior conditioned on every discretized interval, mixing those samples according to $P(X_j)$ or $P(X_j | E_k)$, and applying kernel density estimation (KDE). Conditions are direct death, rescue-delay death, and any life loss. A larger TV distance identifies stronger inference-based association with the selected mode, not causal attribution.



5. Conclusion

- **Consequence-aware eBN construction framework.** Establishes a unified SRM–BN interface in which component CPTs connect continuous black-box models to the discrete BN and consequence-weighted risks are made explicit.
- **Joint risk-informed implementation.** Feeds risk information back to co-adapt input-variable discretization and component-CPT construction, while allocating high-fidelity samples to the limit-state regions governing system reliability.
- Future work.** Scale the implementation to larger eBNs under realistic computational limits, and extend it to systems requiring continuous intermediate or output nodes.

References
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